



Part 3: What is a Data Type? (Full Guide)

Simple Study Notes with Zero Hard Words

1. What is a Data Type?

A **Data Type** is a simple rule that tells the computer three things about your memory box:

- What kind of information you are putting inside it (Text? Numbers? A true/false switch?).
- How much memory space the computer needs to save for it.
- What tasks you are allowed to perform with it (You can calculate math with numbers, but you can't calculate math with words!).

Quick Code Trick: You can print `variable.runtimeType` to ask the computer exactly what data type your box is using.

Data Type	What it holds	Quick Example	Rule to Remember
String	Text, Letters, Emojis	"Hello"	Must always use quotes " "
int	Whole Numbers	25, -5, 0	No dots, decimals, or fractions allowed
double	Decimal Numbers	5.9, 3.14, 5.0	Always has a decimal dot inside
bool	True / False Switch	true or false	Only holds exactly two choices

2. Deep Dive: The 4 Big Data Types

1 String (Text Data)

Stores any combination of letters, words, symbols, or spaces.

```
void main() {  
    String name = "John Doe";  
    String city = "New York";  
    String email = "john@email.com";  
    String greeting = "Hello, World! 🙌";  
    String price = "Price: \"$99.99"; // The \ is only needed if you want to print a real p  
}
```

2 int (Whole Numbers)

Stores whole values. Can be positive, negative, or zero.

```
void main() {  
    int age = 25;  
    int temperature = -5;  
    int score = 0;  
}
```

3 double (Decimal Numbers)

Stores precise numbers with dots. It can also read large scientific numbers or whole values written with a dot.

```
void main() {  
    double height = 5.9;  
    double pi = 3.14159;  
    double whole = 5.0; // Allowed!  
}
```

4 bool (Boolean)

A simple true or false switch. Perfect for checklist options or question comparisons.

```
void main() {  
    bool isStudent = true;  
    bool isGraduated = false;  
  
    int age = 20;  
    bool isAdult = age >= 18; // Computer saves this as 'true'  
}
```

3. Text Tools & Operations (String Methods)

```
void main() {  
    String firstName = "John";  
    String lastName = "Doe";  
  
    // 1. Joining text together (Concatenation)  
    String fullName = firstName + " " + lastName; // "John Doe"  
  
    // 2. Dropping a variable inside a sentence (String Interpolation)  
    // No slashes or backslashes needed! Just use the clean dollar sign $  
    print("My name is $firstName."); // Output: My name is John.  
  
    // 3. Math expressions inside a sentence use curly brackets  
    print("The sum is ${5 + 3}"); // Output: The sum is 8  
}
```



Useful String Tools:

- `s.length`: Tells you how many letters are inside. ("Hello".length is 5).
- `s.toUpperCase()` / `s.toLowerCase()`: Changes text to all BIG or all small letters.
- `s.startsWith("Dart")` / `s.endsWith("ming")`: Checks how the text starts or finishes. Returns true or false.
- `s.contains("Dart")`: Checks if a word is hidden anywhere inside the text sentence.
- `s.split(",")`: Chops a single sentence into a list of separate pieces at the mark. ("A,B".split(",") becomes [A, B]).
- `s.substring(0, 4)`: Chops out a small piece from position 0 to 4.
- `s.trim()`: Safely deletes accidental blank spaces at the very front or very back of the text.

4. Number Tools & Operations (int & double Methods)



int Tools:

- `number.abs()` (Absolute Value): Ignores any negative minus signs. Tells you how far the number is from zero. (-25.abs() becomes 25).
- `number.isEven` / `number.isOdd`: Asks the computer if the number is even or odd. Returns true or false.



double Tools:

- `x.abs()`: Removes the minus sign from decimal numbers (-7.5.abs() becomes 7.5).
- `x.floor()`: Forces the decimal down to the smaller whole number (3.7 becomes 3).
- `x.ceil()`: Forces the decimal up to the larger whole number (3.7 becomes 4).
- `x.round()`: Rounds to the nearest whole neighbor (3.7 becomes 4).
- `x.truncate()`: Simply cuts off the decimal parts and leaves the rest (3.7 becomes 3).
- `x.toInt()`: Converts a decimal number into a clean whole number integer.
- `x.toString()`: Turns the number into text words so you can drop it inside a sentence text box.
- `x.remainder(2)`: Finds what is left over after dividing by a number.